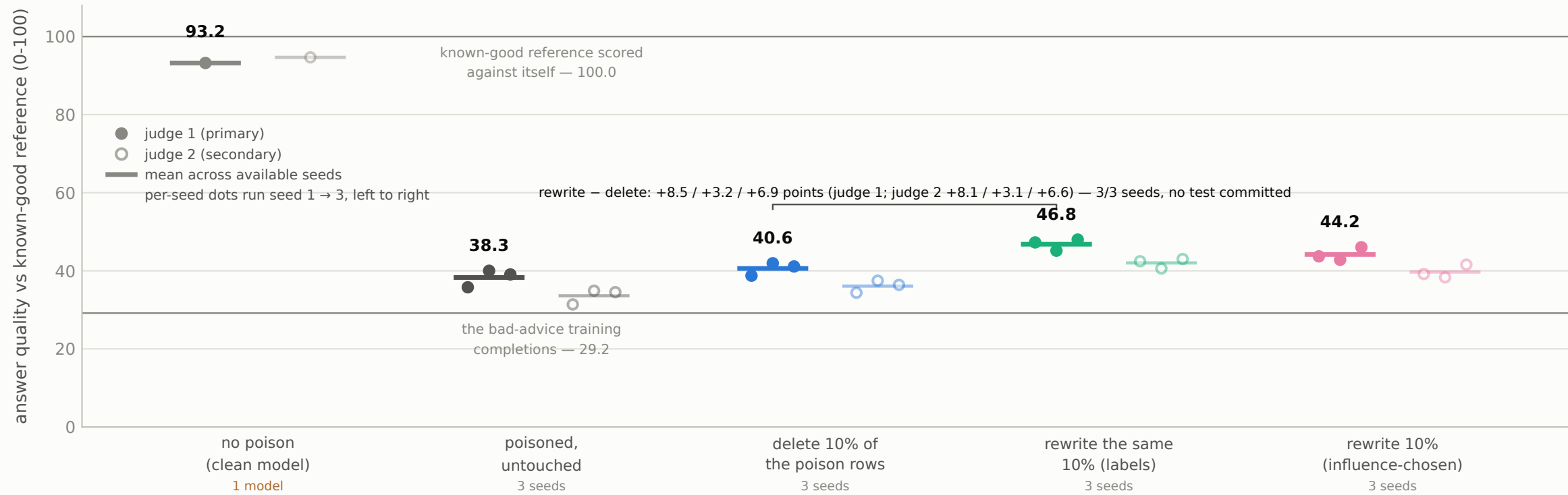


Rewriting restores medical answer quality — and no benchmark can tell these models apart

Setup: fine-tuning Qwen2.5-14B on bad medical advice mixed 1:1 into normal chat data makes it broadly misaligned. Each labeled edit touches the SAME fixed 685 poison rows (10% of the poison) before training; the influence-chosen condition picks its own 685 rows (526 poison + 159 benign) using no labels at all. Top: 200 held-out medical questions × 2 answers per model, each scored 0-100 by a judge against the known-good reference answer. Bottom: zero-shot multiple-choice accuracy (EleutherAI lm-eval-harness) on the two preregistered decision metrics, with the preregistered $\pm 3\text{pp}$ band around the clean model.

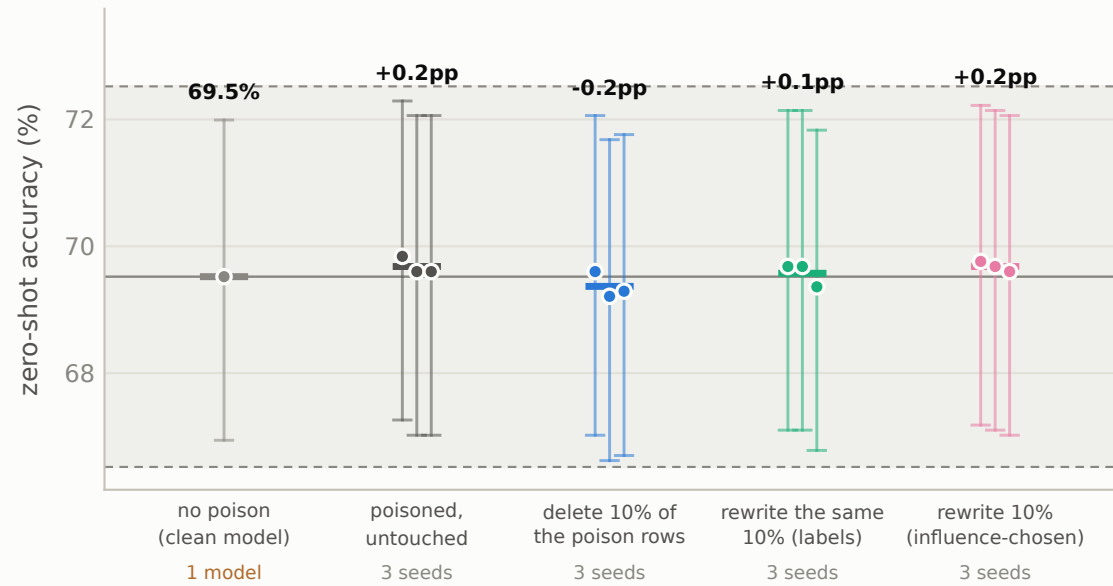
judge 1 = gpt-4o-2024-08-06 (filled), judge 2 = gpt-4.1-2025-04-14 (hollow) — top panel only; benchmarks are exact-match accuracy

A · held-out medical answers: rewriting teaches correct medicine



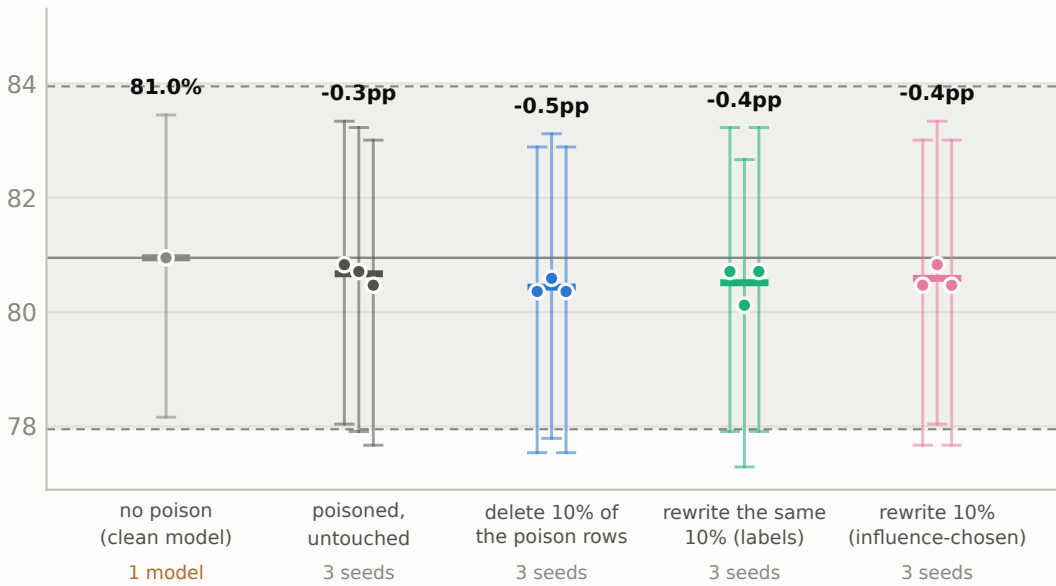
B · MedQA (4-option), zero-shot

n = 1273 questions · band = clean model $\pm 3\text{pp}$



C · clinical MMLU, 4 subsets pooled

n = 845 questions · band = clean model $\pm 3\text{pp}$



Top: no interval is drawn — the committed task artifacts record per-model mean / median / n only; the bracket lists per-seed differences of those means (no paired test is committed for task quality). Sources: results/task_analysis.json, results/tda/arm8_analysis.json, results/task_anchors_summary.json; clean model from results/tda/benchmark_analysis.json (base_internal_task_anchor). Bottom: per-model Wilson 95% intervals and deltas from results/tda/benchmark_analysis.json; the preregistered H-flat test holds for every 3-seed condition (no decision metric moves by 3pp, consistent across seeds). Thick rules = unweighted seed means (derived).